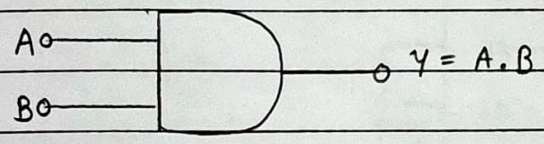
Practical - 1

A. Defination :

Demonstrate the working of basic logic gates and verify with the truth table.

IC :

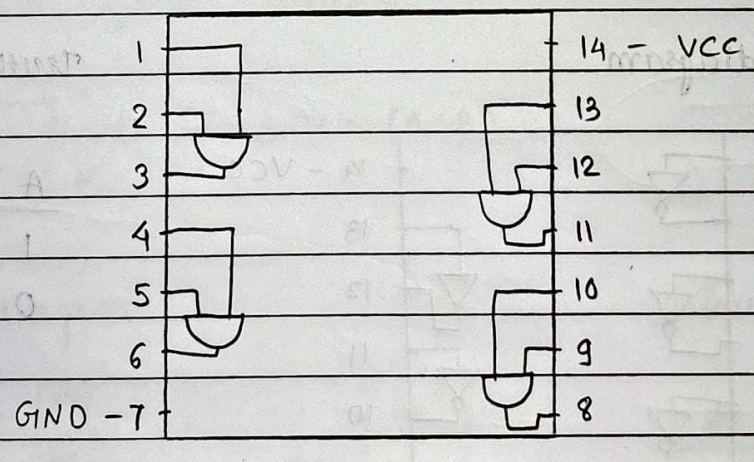
★ AND gate : 7408



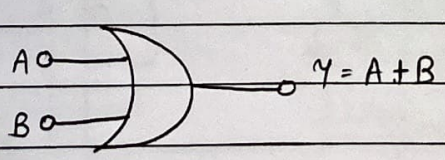
Truth Table

A	B	Y
1	1	1
1	0	0
0	1	0
0	0	0

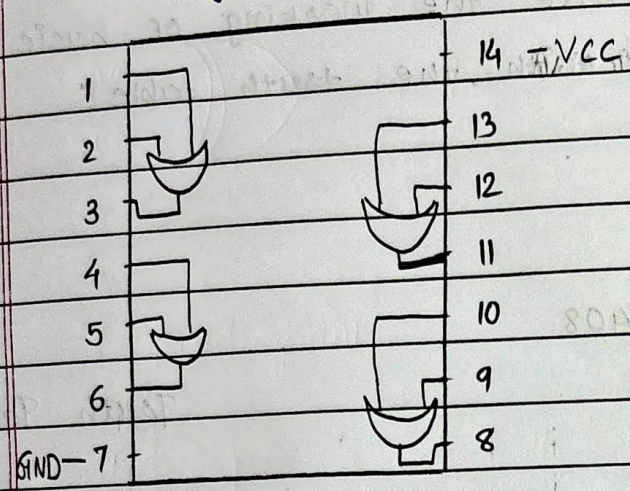
pindiagram :



★ OR gate : 7432



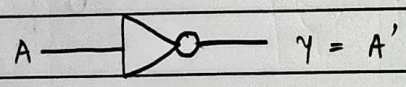
pin diagram:



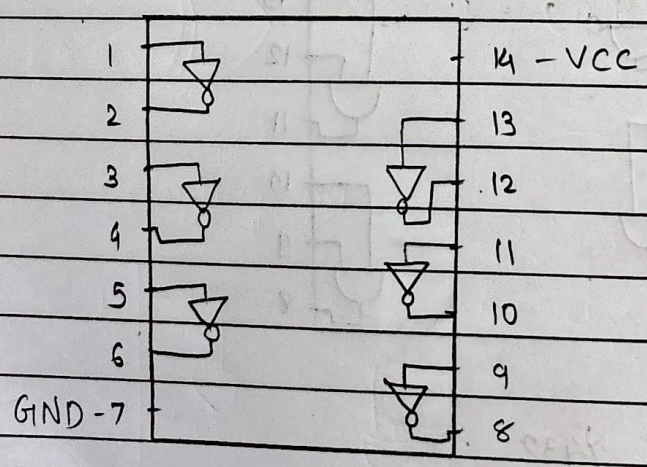
Truth Table

A	B	Y
1	1	1
1	0	1
0	1	1
0	0	0

★ NOT gate : 7404



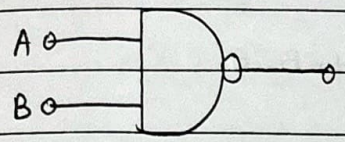
Pin diagram



Truth Table

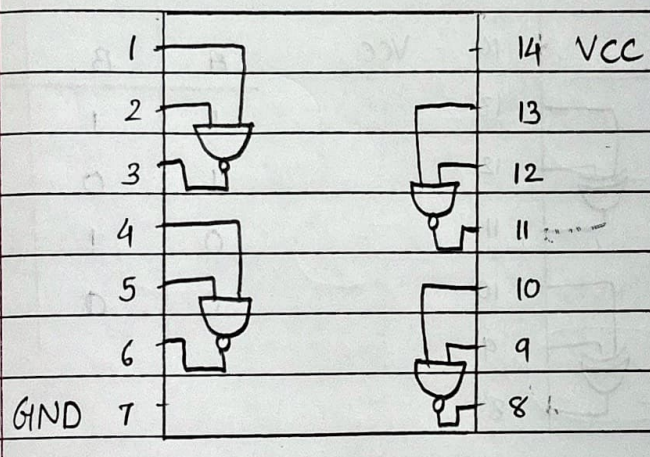
A	Y
1	0
0	1

★ NAND gate: 7400



$Y = (A.B)'$

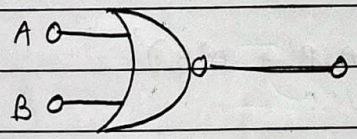
pin diagram



Truth Table

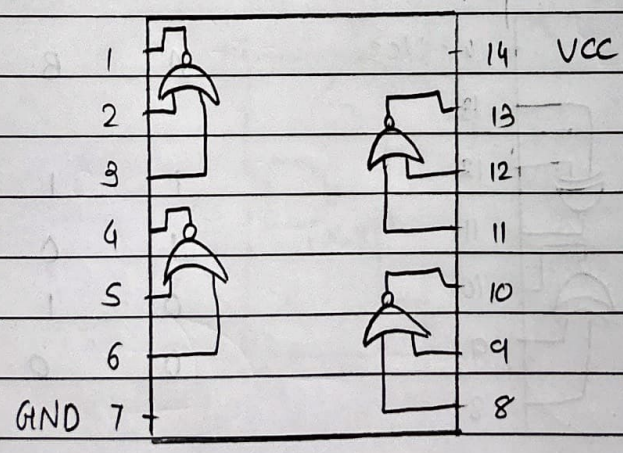
A	B	Y
1	1	0
1	0	1
0	1	1
0	0	1

★ NOR gate: 7402



$Y = (A+B)'$

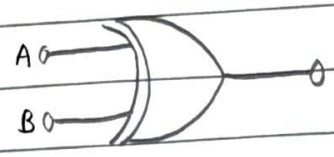
pin diagram



Truth Table

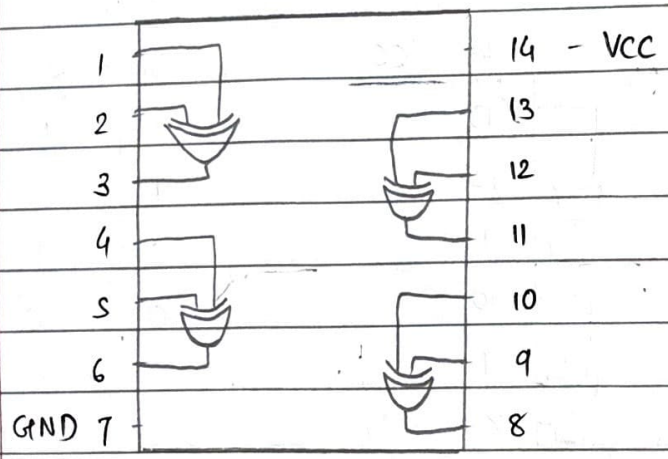
A	B	Y
1	0	0
1	1	0
0	0	1
0	1	0

★ XOR gate: 7486



$Y = A.B' + B.A'$

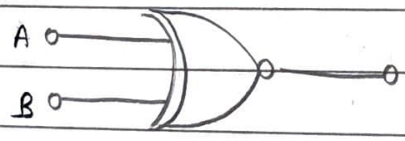
pindiagram



Truth Table

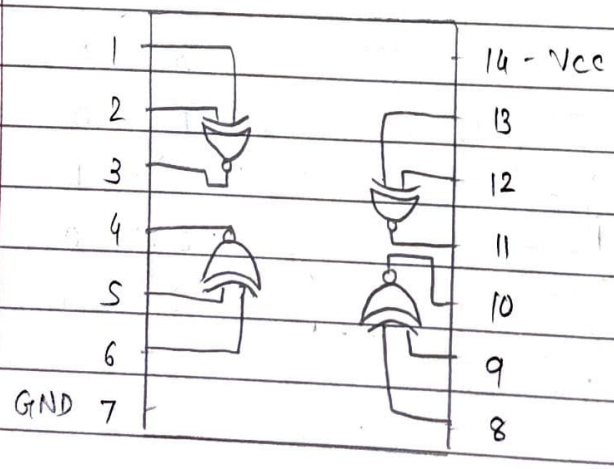
A	B	Y
1	1	1
1	0	0
0	1	0
0	0	1

★ XNOR gate : 74266



$Y = (A.B' + A'B)'$

Pindiagram



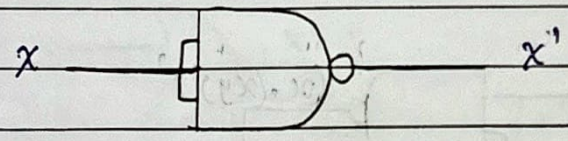
Truth Table

A	B	Y
1	1	0
1	0	1
0	1	1
0	0	0

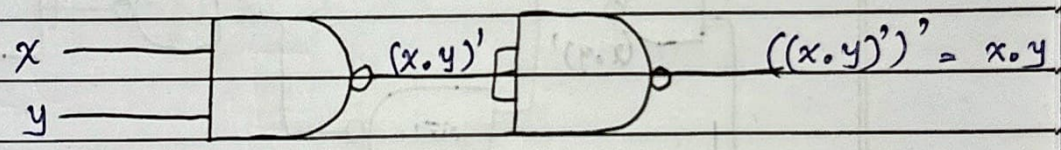
B. Demonstrate the working of basic logic gates using NAND and NOR gates

⇒ Using NAND gate:

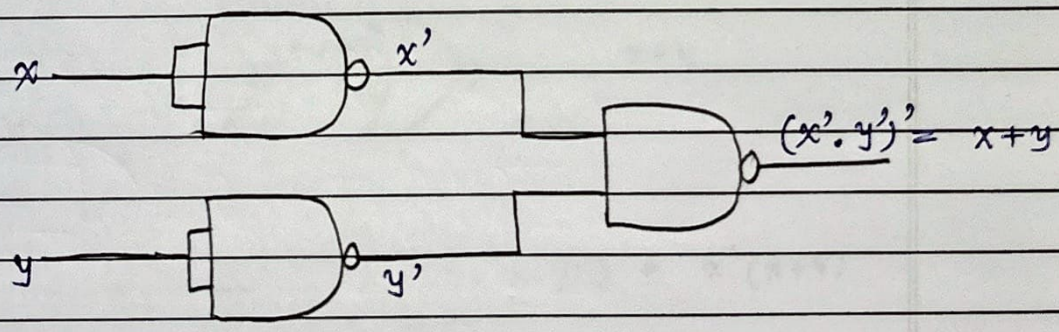
NOT :



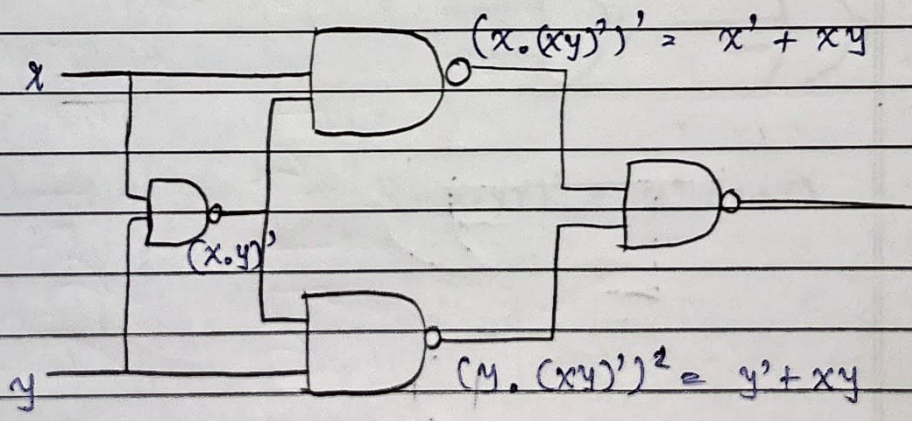
AND :



OR :



XOR :



output : $((x'+xy).(y'+xy))'$

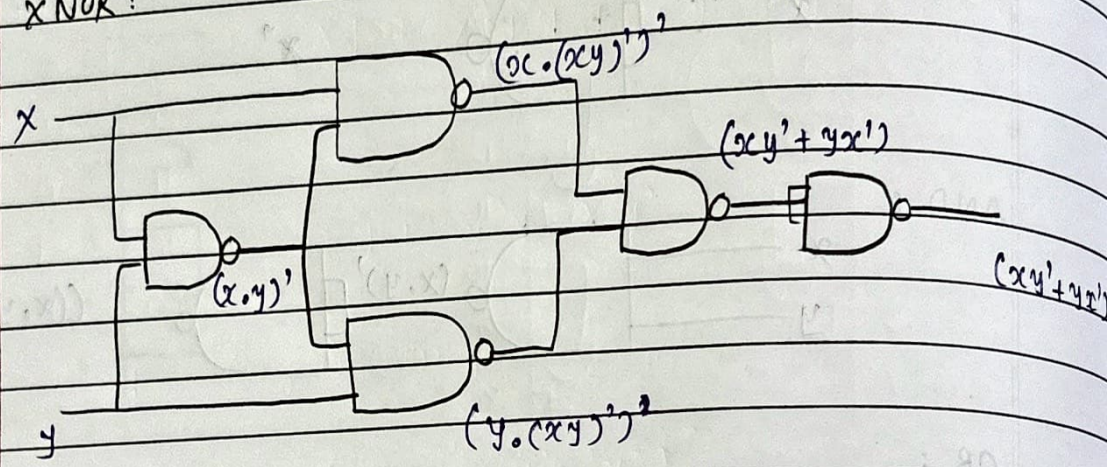
$(x'+xy)' + (y'+xy)'$

$$= x.(x'+y') + y.(x'+y')$$

$$= \underbrace{xx'}_{0} + xy' + yx' + \underbrace{yy'}_{0}$$

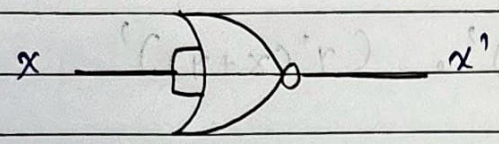
$$= xy' + yx'$$

XNOR:

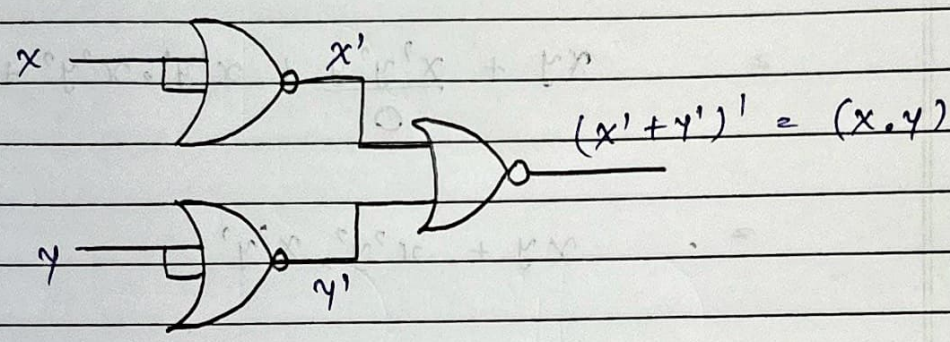


⇒ Using NOR gate:

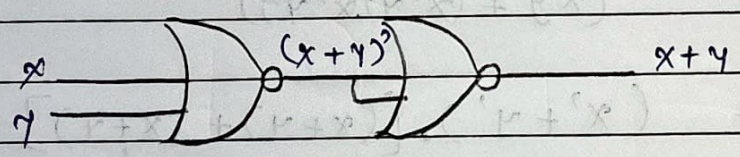
NOT :



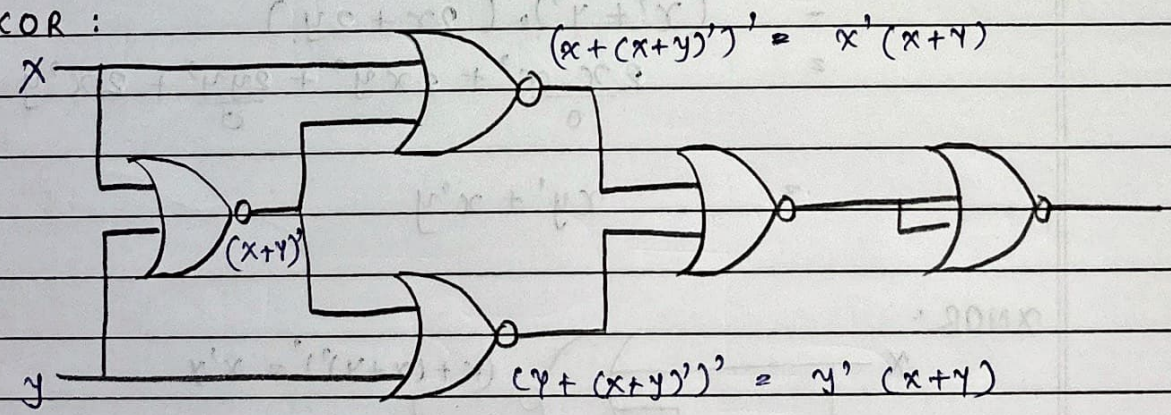
AND :



OR :



XOR :



output :

$$= (x'(x+y) + y'(x+y))'$$

$$= (x'(x+y))' \cdot (y'(x+y))'$$

$$= (x + (x'y')) \cdot (y + (y'x'))$$

$$= xy + \underbrace{x'y'y}_{0} + x'y' \cdot x'y' + \underbrace{x'y'x}_{0}$$

$$= xy + x'y' \cdot x'y'$$

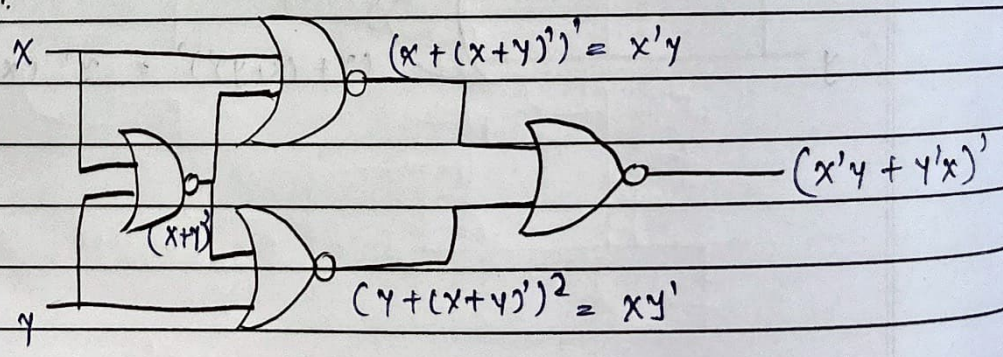
by complimenting

$$= (xy + (x'y')(x'y'))'$$

$$= (x'+y') \cdot [(x+y) + (x+y)]$$

$$= (x'+y') \cdot (x+y)$$

$$= \underbrace{x \cdot x'}_{0} + xy' + \underbrace{yy'}_{0} + x'y$$

$$= xy' + x'y$$

XNOR :